

Local Restoring Strength in the Kagome XY Model: An Energetic Descriptor of Vortex Dynamics

Abstract

We introduce the local restoring strength (drop), defined as the difference in bond energy between the vortex core ($r=0$) and its nearest-neighbor shell ($r=1$) in the kagome XY model, and investigate its role as an intermediate observable connecting the energetic and dynamical descriptions of vortex formation.

Through Monte Carlo simulations at $L=128$ ($K=0.7$, 63 temperature points, 5 seeds), we find that drop correlates strongly with the mean bond energy, vortex birth rate, total vortex count, specific heat, and helicity modulus across the full temperature range studied, indicating that drop functions as a thermodynamic probe that simultaneously reflects the energetic and dynamical state of the system. Transfer entropy analysis confirms that the energy-to-winding-number direction ($E \rightarrow W$) dominates in 81% of temperature points, supporting a directional ordering $\text{Energy} \rightarrow \text{drop} \rightarrow \text{Birth} \rightarrow N_{\text{vortex}}$.

A key negative result strengthens the interpretation: no measurable correlation is found between drop at individual vortex sites and the local information density, even when the information window is reduced to the vortex-core scale ($R_{\text{LOCAL}}=1$, $r=-0.014$). This indicates that information is not encoded by individual vortices but rather emerges from their collective organization — consistent with the companion study (Paper I).

Furthermore, the residual of the linear fit $\text{drop} \approx a \cdot \text{Energy} + b$ ($R^2=0.995$) shows no significant correlation with birth rate, specific heat, or helicity modulus ($|r|<0.13$, all $p>0.32$), indicating that drop is primarily an energetic descriptor with little independent dynamical contribution beyond the bond energy.

1. Introduction

The Berezinskii-Kosterlitz-Thouless (BKT) transition [1,2] describes the unbinding of topological defect pairs in two-dimensional systems without conventional symmetry breaking. A companion study (Paper I) demonstrated that the BKT transition of the kagome XY model leaves distinct signatures in the spatial information structure of the lattice, identifying a sharp peak in the vortex-emergence efficiency $\eta = \text{birth}/\sqrt{MI}$ that is independent of system size across $L=48\text{--}256$ [4]. However, the microscopic energetic mechanism that drives vortex formation remained an open question.

The present work addresses this question by introducing the local restoring strength (drop), defined as the difference in mean bond energy between the vortex core plaquette ($r=0$) and its nearest-neighbor shell ($r=1$):

$$\text{drop} \equiv E(r=0) - E(r=1)$$

This quantity measures how much more energetically excited the immediate environment of a vortex is compared to its surroundings. We find strong evidence that drop serves as an intermediate energetic descriptor bridging the microscopic energy structure and the macroscopic vortex dynamics, while the relation between drop and the collective information field remains to be clarified.

At the macroscopic level, drop is almost perfectly correlated with the bond energy, vortex birth rate, vortex count, specific heat, and helicity modulus. Transfer entropy analysis confirms the directional ordering $\text{Energy} \rightarrow \text{drop} \rightarrow \text{Birth} \rightarrow N_{\text{vortex}}$. At the microscopic level, no measurable correlation exists between drop at individual vortex sites and the local information density at any tested spatial scale — establishing the irreducibly collective character of information in this system.

The paper is organized as follows. Section 2 defines the model, observable drop, and analysis methods. Section 3 presents the results. Section 4 discusses the physical interpretation and connection to Paper I. Section 5 concludes.

2. Model and Methods

2.1 Kagome Lattice XY Model

We study the XY model on the kagome lattice with Hamiltonian $H = -K \sum \langle ij \rangle \cos(\phi_i - \phi_j)$, $K=0.7$, on $N=3L^2$ sites. All simulations use the Metropolis algorithm with $N_{\text{therm}}=200$, $N_{\text{meas}}=60$, at $L=128$, $T=0.75\text{--}2.00$ ($\Delta T=0.02$, 63 points), 5 seeds. The BKT transition temperature is $T_{\text{BKT}} \approx 0.76$ at this system size.

2.2 Local Restoring Strength (drop)

For each vortex at plaquette p , we compute $E(r)$ via breadth-first search on the plaquette adjacency graph. $\text{drop}_i \equiv E(r=0) - E(r=1)$ at the individual vortex level; $\text{drop}(T)$ is the ensemble average over all vortices at temperature T (Figure 1).

2.3 Information-Theoretic Observables

The local Shannon entropy $H(ix, iy) = -\sum_w p(w) \log_2 p(w)$ is computed from plaquette winding number distributions within a spatial window of radius R_{LOCAL} . $I = H_{\text{max}} - H$ ($H_{\text{max}} = \log_2 8 = 3$), and $\sqrt{MI} = \sqrt{\langle I^2 \rangle}$. We test $R_{\text{LOCAL}} = 3$ and $R_{\text{LOCAL}} = 1$ (vortex-core scale).

2.4 Causal Analysis

Transfer Entropy $TE(X \rightarrow Y) = H(Y_t | Y_{t-1}) - H(Y_t | Y_{t-1}, X_{t-1})$ is computed with a discrete-bin estimator (6 bins, lag=1 sweep). Net TE = $TE(E \rightarrow W) - TE(W \rightarrow E)$ quantifies the dominant causal direction.

2.5 Residual Decomposition

We fit $\text{drop}(T) = a \cdot \text{energy}(T) + b$ and define the residual $\delta \text{drop}(T) = \text{drop}(T) - \text{fit}(T)$, testing whether δdrop correlates with birth rate, C_v , N_{vortex} , or Y .

3. Results

3.1 The r-Profile: Physical Basis for drop

Figure 1 shows the radial energy profile $E(r)$ for representative temperatures. Bond energy at the vortex core ($r=0$) is substantially elevated above bulk equilibrium and recovers monotonically within $r=2-3$ plaquette shells, motivating $E(r=1)$ as the near-field reference. Drop(T) decreases monotonically from ≈ 0.64 at $T=0.75$ to ≈ 0.42 at $T=2.00$.

Figure 1: Definition and temperature dependence of the local restoring strength (drop)

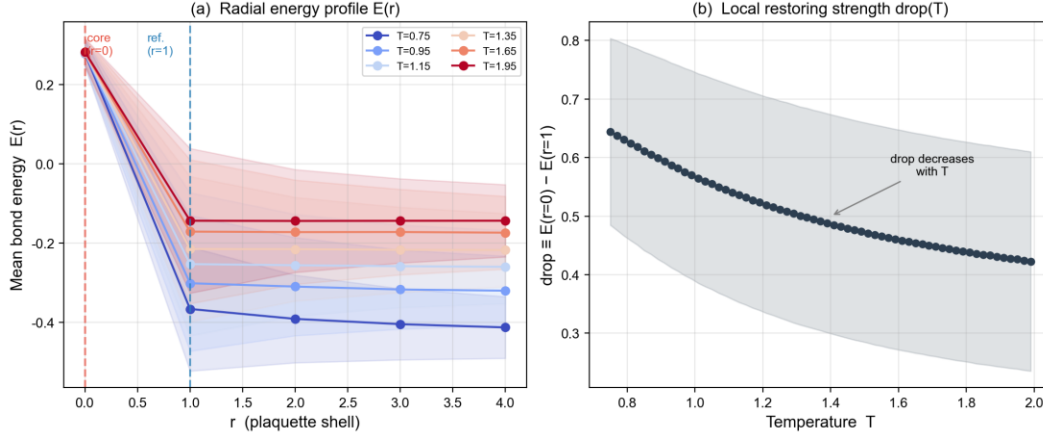


Figure 1. (a) Radial energy profile $E(r)$ for representative temperatures. Bond energy is elevated at the vortex core ($r=0$, red dashed) and recovers to equilibrium within $r=2-3$ shells. (b) Temperature dependence of $\text{drop}(T) \equiv E(r=0) - E(r=1)$, the local restoring strength.

3.2 Macroscopic Correlations

Figure 2 shows $\text{drop}(T)$ alongside energy and birth rate. Table 1 summarizes all macroscopic correlations. Drop is almost perfectly correlated with energy ($r=-0.998$, $R^2=0.995$), birth rate ($r=-0.998$, $R^2=0.996$), and vortex count ($r=-1.000$, $R^2=0.999$). The correlation with N_{vortex} ($R^2=0.999$) is particularly striking, suggesting that drop and vortex count are essentially functionally related across the entire temperature range.

Figure 2: Temperature dependence of drop, energy, and birth rate ($L=128$)

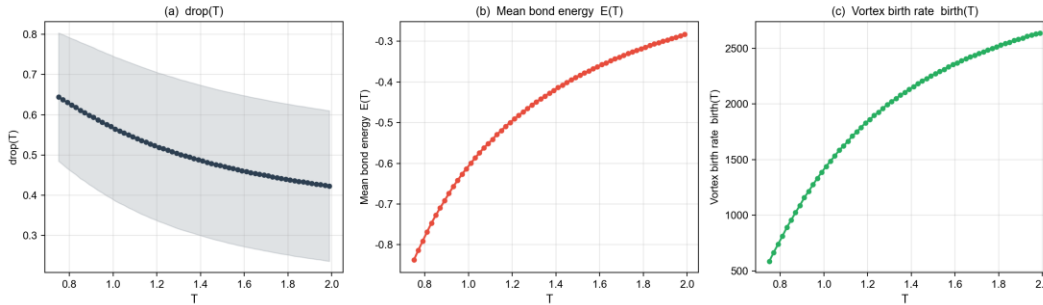


Figure 2. Temperature dependence of (a) drop, (b) mean bond energy, and (c) vortex birth rate ($L=128$). All three quantities decrease monotonically with temperature and track each other closely.

Table 1. Macroscopic correlations with $\text{drop}(T)$ ($L=128$, $n=63$ T-points)

Energy: $r = -0.998$, $R^2 = 0.995$, $p = 1.4 \times 10^{-72}$

Birth rate: $r = -0.998$, $R^2 = 0.996$, $p = 2.2 \times 10^{-75}$
N_vortex: $r = -1.000$, $R^2 = 0.999$, $p = 6.8 \times 10^{-96}$
Cv: $r = +0.970$, $R^2 = 0.941$, $p = 3.5 \times 10^{-39}$
Y (helicity): $r = +0.840$, $R^2 = 0.706$, $p = 7.2 \times 10^{-18}$

Figure 3: Macroscopic correlations between drop and thermodynamic observables (L=128)

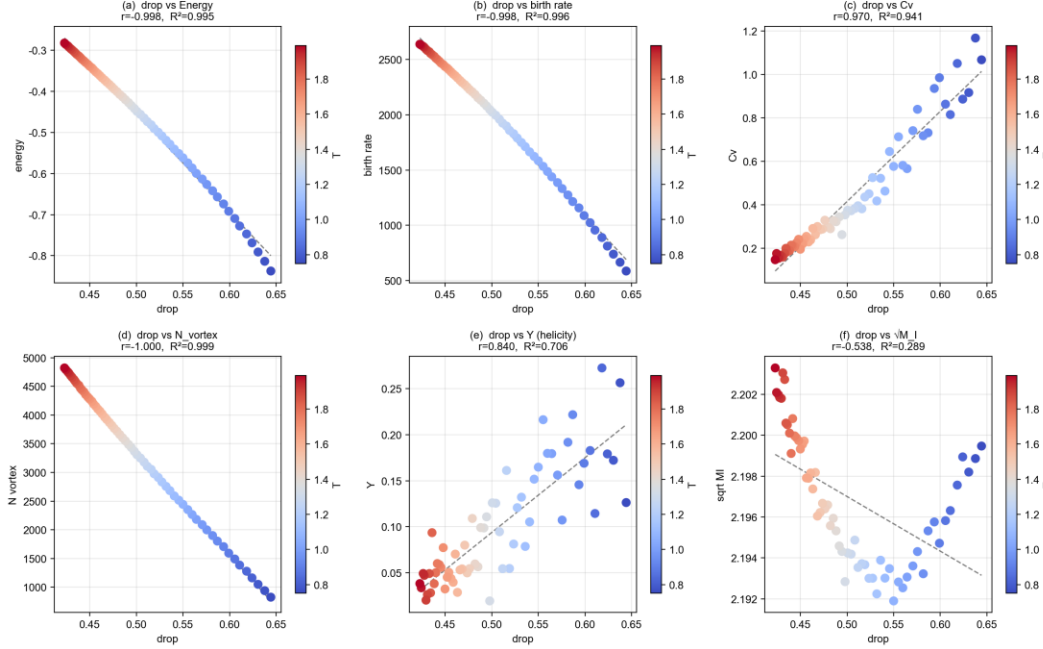


Figure 3. Scatter plots of drop versus macroscopic observables. Each point represents one temperature T (color-coded). Dashed lines show linear fits. All correlations are statistically significant ($p < 10^{-17}$).

3.3 Causal Analysis

Transfer entropy analysis (Figure 4) shows $TE(E \rightarrow W) > TE(W \rightarrow E)$ in 81% of temperature points (51/63), supporting energy as the primary driver in this system. Note that transfer entropy was evaluated only for the $E \rightarrow W$ link; the correlation strengths reported below for the other links are cross-sectional (across temperature) and are consistent with, but do not independently establish, the proposed Energy $\rightarrow$ drop $\rightarrow$ Birth $\rightarrow$ N_vortex ordering:

E $\rightarrow$ drop: $r = -0.998$, $R^2 = 0.995$
drop $\rightarrow$ birth: $r = -0.998$, $R^2 = 0.996$
birth $\rightarrow$ N: $r = +0.996$, $R^2 = 0.992$
drop $\rightarrow$ N: $r = -1.000$, $R^2 = 0.999$

Figure 4: Transfer entropy analysis — energy drives winding number ($L=128$)

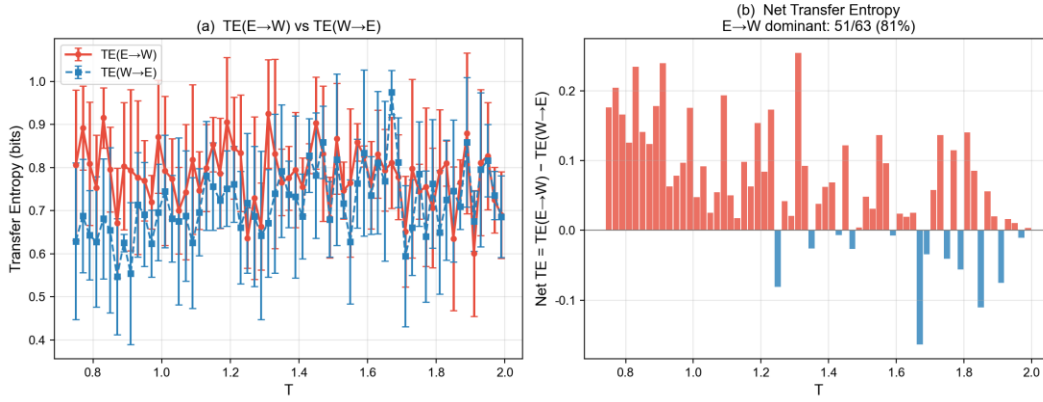


Figure 4. Transfer entropy analysis. (a) $TE(E \rightarrow W)$ consistently exceeds $TE(W \rightarrow E)$ across the temperature range. (b) Net $TE = TE(E \rightarrow W) - TE(W \rightarrow E)$ is positive in 81% of temperature points, confirming the energy-to-winding-number causal direction.

3.4 Residual Analysis

The linear fit $\text{drop}(T) = a \cdot \text{energy}(T) + b$ achieves $R^2=0.995$. The residual δdrop shows no significant correlation with any other observable: birth rate ($r=-0.008$, $p=0.95$), C_v ($r=-0.057$, $p=0.66$), N_{vortex} ($r=-0.093$, $p=0.47$), Y ($r=+0.127$, $p=0.32$). Drop is primarily an energetic descriptor with little independent dynamical contribution beyond the bond energy.

3.5 Absence of Local Drop–Information Correlation

Figure 5 shows the central negative result. At $R_{\text{LOCAL}}=3$: $r \approx 0.001$ ($n=5 \times 10^6$). At $R_{\text{LOCAL}}=1$ (vortex-core scale): $r=-0.014$. Both are statistically indistinguishable from zero. This rules out the spatial-averaging hypothesis and establishes that local information density is not a property of individual vortices at any tested spatial scale.

Figure 5: Absence of local drop–information correlation at any spatial scale

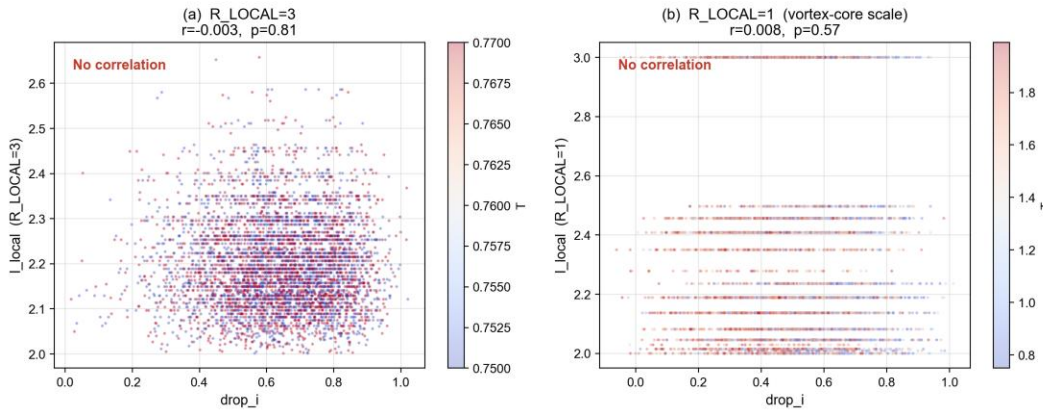


Figure 5. Absence of local drop–information correlation. (a) $R_{\text{LOCAL}}=3$: $r \approx 0.001$. (b) $R_{\text{LOCAL}}=1$ (vortex-core scale): $r=-0.014$. Neither correlation is distinguishable from zero, ruling out spatial averaging as an explanation and establishing the collective character of information.

4. Discussion

4.1 drop as a Local Energetic Descriptor

The results establish drop as a physically interpretable local energetic descriptor that mediates between the microscopic energy structure of the lattice and its macroscopic vortex dynamics. The near-perfect correlations ($R^2=0.992-0.999$) between drop and energy, birth rate, vortex count, and specific heat indicate that drop is a local projection of the same thermodynamic state that governs these global quantities. The Transfer Entropy directional dominance ($E \rightarrow W$ in 81% of temperature points) is consistent with energy as the primary driver, propagating through drop to determine vortex dynamics.

The residual analysis confirms this picture: once the energetic component is removed, no residual correlation with topological or dynamical observables remains. Drop carries no independent topological content beyond what is already encoded in the bond energy.

4.2 Information as an Emergent Collective Property

The absence of a local drop–information correlation at any tested spatial scale indicates that information is not encoded by individual vortices but emerges from their collective organization. This mirrors statistical mechanics, where entropy is an ensemble property rather than a property of a single particle.

This conclusion connects naturally to Paper I, which found that $\sqrt{MI}$ exhibits a sharp collective signature near the BKT transition that is system-size-independent across $L=48-256$. The present study shows that this collective signal cannot be reduced to any single-vortex property. Paper I asked ‘what does the information structure look like?’ and found an answer at the collective level. Paper II asks ‘where does it come from?’ and finds that individual vortices, characterized by drop, do not carry the answer — only their collective organization does.

4.3 A Two-Level Description of the BKT Transition

The combined results suggest a two-level description:

Table 2 summarizes the two-level structure identified in this study.

See Table 2: Local level (drop) — individual vortex, energetic, local bond; Collective level ($\sqrt{MI}$) — vortex ensemble, informational, global spatial organization.

The gap between these levels is precisely the negative result of Section 3.5: there is no single-vortex observable that bridges local energetics and collective information.

4.4 Temperature-Dependent Coupling $\alpha(T)$

The proportionality coefficient $\alpha(T) = \text{drop}/C_v$ increases monotonically with temperature ($C_v=0.43$), indicating that the coupling between local restoring strength and thermal fluctuations is not a fixed constant but varies with thermodynamic state. The functional form of $\alpha(T)$ and its connection to BKT RG flow equations are left for future investigation. An intriguing numerical coincidence between drop and the Landauer energy scale ($k_B T \ln 2$) near $T \approx 0.87$ is reserved for future investigation.

4.5 Temporal Ordering and the Slow Dynamics of Information

To test whether the transfer-entropy directionality identified in Section 3.3 ($E \rightarrow W$) extends to a full causal ordering among all observables, we performed an additional equilibrium (undriven) analysis

recording $E(t)$, $\text{drop}(t)$, $\text{birth}(t)$, and $\sqrt{MI}(t)$ simultaneously at every Monte Carlo sweep across the same 63 temperature points (5 seeds each, $N_{\text{meas}}=2000$ sweeps per run — substantially longer than the $N_{\text{meas}}=60$ used elsewhere in this study, since transfer-entropy estimation requires longer time series than the macroscopic T-dependence analysis of Section 3). Net transfer entropy and cross-correlation lag were computed for all six pairs among these four observables.

The $E \rightarrow \text{drop}$ and $E \rightarrow \text{birth}$ directions were confirmed as dominant, with net transfer entropy excluding zero (95% CI across seeds) in 70% and 87% of temperature points respectively, consistent with the Energy $\rightarrow$ drop $\rightarrow$ Birth chain established above. Net transfer entropy also indicated that $\sqrt{MI}$ statistically precedes E , drop, and birth in 65–94% of temperature points. However, measuring the autocorrelation time (Sokal estimator) of each observable revealed $\tau_{\text{auto}}(\sqrt{MI}) \approx 147$ sweeps, compared to $\tau_{\text{auto}}(E) \approx 2.8$, $\tau_{\text{auto}}(\text{drop}) \approx 0.64$, and $\tau_{\text{auto}}(\text{birth}) \approx 0.65$ sweeps — a separation of roughly two orders of magnitude. Because transfer entropy is known to favor slowly-varying (long-memory) variables independent of any genuine causal role, we do not interpret the apparent precedence of $\sqrt{MI}$ as evidence of causal ordering.

What this analysis robustly establishes is that the collective information field $\sqrt{MI}$ evolves, in equilibrium, on a timescale approximately two orders of magnitude slower than the local energetic and topological observables (E , drop, birth). The physical origin of this separation of timescales — including its dependence on temperature and system size — is left for future investigation.

This result highlights an important methodological point: transfer-entropy analyses applied to observables with widely separated autocorrelation times should be interpreted cautiously, since long-memory variables may appear to statistically precede faster-varying variables even in the absence of any direct causal influence.

5. Conclusions

We have introduced the local restoring strength (drop) as a local energetic descriptor of individual vortices in the kagome XY model. Figure 6 summarizes the key results.

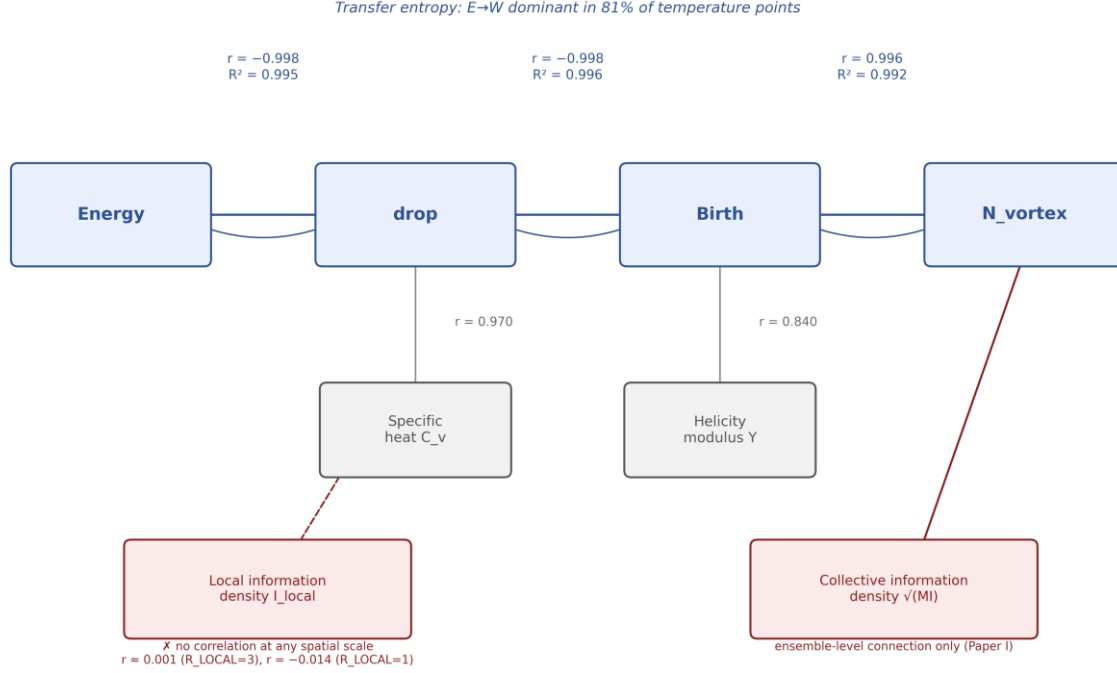


Figure 6. Summary diagram. Directional structure Energy → drop → Birth → N_vortex, with key statistics at each link (top), and drop's secondary correlations with specific heat and helicity modulus (middle). Transfer entropy confirms E→W dominance in 81% of temperature points. The collective information $v(MI)$ connects only at the ensemble level; no correlation exists between individual-vortex drop and local information density (X , dashed). See Table 3 for the corresponding numerical values.

Table 3. Key numerical results for the causal chain and information decoupling ($L=128$, $K=0.7$, $T=0.75-2.00$, $\Delta T=0.02$, 63 points, 5 seeds).

Relationship	r	R^2
drop vs. mean bond energy (E)	-0.998	0.995
drop vs. vortex birth rate	-0.998	0.996
drop vs. total vortex count (N_vortex)	-1.000	0.999
drop vs. specific heat (C_v)	0.970	—
drop vs. helicity modulus (Y)	0.840	—
Transfer entropy: E→W dominant	81% of T-points	—
drop_i vs. local information ($R_{\text{LOCAL}}=1$)	-0.014	—
drop_i vs. local information ($R_{\text{LOCAL}}=3$)	≈ 0.001	—

1. drop is almost perfectly correlated with mean bond energy ($r=-0.998$, $R^2=0.995$), vortex birth rate ($r=-0.998$, $R^2=0.996$), total vortex count ($r=-1.000$, $R^2=0.999$), specific heat ($r=0.970$), and helicity modulus ($r=0.840$).
2. Transfer entropy establishes $E \rightarrow W$ as the dominant directional channel in 81% of temperature points, supporting Energy $\rightarrow$ drop $\rightarrow$ Birth $\rightarrow$ N_vortex.
3. Residual analysis indicates that drop is primarily an energetic descriptor: $R^2=0.995$ for the linear fit with energy, and the residual carries no significant correlation with any topological or dynamical observable examined here.
4. No measurable correlation exists between drop_i and local information density at any spatial scale ($R_LOCAL=1$: $r=-0.014$; $R_LOCAL=3$: $r \approx 0.001$), ruling out spatial averaging and establishing the collective character of information.
5. Within the observables examined here, the collective information field $\sqrt{MI}$ cannot be reduced to any single-vortex energetic descriptor (drop), extending the collective picture of Paper I.
6. In equilibrium, the collective information field $\sqrt{MI}$ evolves on a timescale approximately two orders of magnitude slower than the local energetic and topological observables ($\tau_auto(\sqrt{MI}) \approx 147$ sweeps vs. $\approx 1-3$ sweeps for E , drop, and birth). A preliminary transfer-entropy analysis suggested that $\sqrt{MI}$ statistically precedes E , drop, and birth; however, this separation of autocorrelation times indicates the apparent precedence more likely reflects a known long-memory bias of transfer entropy than genuine causal ordering, and no causal claim is made here. The physical origin of this slow collective dynamics is left for future investigation (Section 4.5).

Together, these results suggest that energetic observables and collective information characterize complementary levels of description of the BKT transition, with drop capturing the local energetic response of individual vortices and $\sqrt{MI}$ capturing the collective information structure that emerges from their ensemble.

Acknowledgments

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With my deepest gratitude, I dedicate this work to my father-in-law, In-sang, Ryu.

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